

BIO203,BIO301,BIO302,ZOO505

Viva preparation 2023 by

SULMAN ALI

BS-ZOOLOGY

1= DNA Extraction:

DNA extraction is a method to purify DNA by using physical and/or chemical methods from a sample separating DNA from cell membranes, proteins, and other cellular components.

Why we do DNA Extraction: To DNA Profiling, to separate genetic material from the rest of the cell (proteins, RNA, cell membrane, etc.), to test a newborn for a genetic disease, analyze forensic evidence, or study a gene involved in cancer.

Application of this Method: To Studying the genetic causes of disease and for the development of diagnostics and drugs. It is also essential for carrying out forensic science, sequencing genomes, detecting bacteria and viruses in the environment and for determining paternity.

Materials Required:

Name:	Blood Sample	Lysis Buffer	A1 Buffer	SDS 10%	Proteinase K	Phenol chloroform isoamyl alcohol (PCM)	Iso-propanol	Ethanol 70%	Evaporate organic compounds:
Quantity:	200ul	800ul	-----	-----	-----	-----	---	--	

Function	To collect DNA	To break down the blood separate the RBCs.	To resuspension of cell pellet for plasmid DNA extraction.	Linearize the proteins into primary, secondary, tertiary structures.	To break down these proteins structures.	To remove salts	Separate upper layer and precipitate the DNA.	To wash-up the extracted DNA and to remove contaminants from it.	To dry-up the DNA at least 4 to 7 hours.
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Method: Add one by one buffers in 200ul blood sample in the micro centrifuge tube. Vortex it for 30 sec and centrifuge it in 15000 rpm for 3 mints. In the same way add all the buffers and reagents one by one with same method. In last wash up the DNA by 70% ethanol and leave it to dry it at least 4 to 7 hours.

2= Gel- Electrophoresis:

It is a laboratory technique used to separate DNA, RNA or protein molecules based on their size and electrical charge, the molecules to be separated are pushed by an electrical field through a gel that contains small pores.

Why we do Gel- Electrophoresis: To separation of DNA fragments for DNA fingerprinting to investigate crime scenes. To analyze results of polymerase chain reaction. To analyze genes associated with a particular illness.

Application of this Method: DNA profiling for taxonomy studies to distinguish different species. To separation of DNA fragments for DNA fingerprinting to investigate crime scenes. To analyze results of polymerase chain reaction. To analyze genes associated with a particular illness.

Materials Required:

1 gram Agarose powder	1x TAE Buffer 500ml	Ethidium bromide
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Procedure: To make a gel, agarose powder is mixed with an electrophoresis buffer and heated to a high temperature until all of the agarose powder has melted. The molten gel is then poured into a gel casting tray and a “comb” is placed at one end to make wells for the sample to be pipetted into. Run the gel by connecting DC voltage(120) supply and in last view this gel under the UV transilluminator.

3= Nanodrop (Spectrophotometer):

The Nanodrop Spectrophotometer is used primarily to measure DNA, RNA or protein concentrations in the UV/visible light range. It can be used to measure absorbance spectra or single wavelengths of these macromolecules in solution.

230 to 280 nm wavelength	Is to agjusted
1.8 to 2 ratio	Pure DNA
Below 1.7	Contamination of phonetic compounds.
Higher to 2.2	Contaminations of proteins

Why we do this method: to check the purity of the extracted DNA or RNA and their quantities and the contaminants in them.

Applications of Nanodrop (Spectrophotometer): to check the purity of the extracted DNA or RNA and their quantities and the contaminants in them.

Procedure: Simply pipet a drop of your DNA, RNA, or protein sample—as little as 1.0–2.0 μL —on the pedestal and pull down the arm. No cuvettes or capillaries are needed and the results appear in seconds. And set the software as we like.

4= Primers Details:

PCR primers are short pieces of single-stranded DNA, usually around 20 nucleotides in length.

Function of primers: to find targeted gene or for locating muted gene.

Annealing temperature of primers	48 to 62 °c
Length	18 to 22 or 30 There may be deference appears in +-2
GC Contents	Almost 60%

5= PCR:

Polymerase chain reaction (abbreviated PCR) is a laboratory technique for rapidly producing (amplifying) millions to billions of copies of a specific segment of DNA, which can then be studied in greater detail.

Why we do PCR: PCR is a common tool used in medical and biological research labs. It is used in the early stages of processing DNA for sequencing?, for detecting the presence or absence of a gene to help identify pathogens? during infection, and when generating forensic DNA profiles from tiny samples of DNA.

Applications of PCR: Testing of genetic disease mutations, Monitoring the gene in gene therapy, Detecting disease-causing genes in the parents.

Material required:

Name	Quantity
DNA	1uL
Primers R-F	Both 1 uL
Taq Polymerase	1 uL
Buffer Taq	2.5 uL
Mgcl ₂	2 uL
dNTPs	2.5 uL
Water	10uL

Time and temperature for Thermocyclers:

Step	Temp	Time	# of cycles
Initial Denaturation	94°C	5 min	30-35
Denaturation	94°C	30 sec	
Primer Annealing	T _m -5°C	45 sec	
Extension	72°C	1 min per kb	
Final Extension	72°C	5 min	

Procedure:

PCR is based on three simple steps required for any DNA synthesis reaction: (1) denaturation of the template into single strands; (2) annealing of primers to each original strand for new strand synthesis; and (3) extension of the new DNA strands from the primers.

6= Definitions and their formulas:

Molarity: Molarity of a given solution is defined as the total number of moles of solute per litre of solution.

$M = \frac{\text{Molecular weight}}{1000 \text{ ml of distilled water}} \Rightarrow 1M$

Normality: Normality (N) is defined as the number of mole equivalents per liter of solution:

$\text{normality} = \frac{\text{number of mole equivalents}}{1 \text{ L of solution}}$

Parts per Millions: This is a way of expressing very dilute concentrations of substances.

$\text{PPM} = \frac{\text{Molecular weight}}{\text{Dissociated ions}}$